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SE Lab – 3

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1. Architecture Selection

For our Campus Placement & Internship Pipeline System, we chose a Layered Architecture because it fits naturally with how our placement workflows and academic operations are structured.

2. Architectural Choice

We selected a Layered Architecture to clearly separate the user interface, our core placement business logic, and the underlying database storage. This clean separation makes it straightforward to build, test, and maintain components like our resume parsing engine, multi-tier eligibility verification rules, interview scheduling queues, and digital offer generation workflow.

i) Reason 1 — Clean Separation of Duties

Our platform handles multiple distinct operations—ranging from students uploading resumes and officers setting up corporate drives, to tracking live interview rounds and managing candidate offer responses. Splitting these responsibilities into horizontal layers keeps our code organized and simple to manage.

ii) Reason 2 — Robust Data Security and Management

Because the portal manages sensitive student data, including academic records, CGPA profiles, active backlogs, and confidential offers, we needed a structured way to enforce strict security controls. A layered model allows us to implement AES-256 encryption at rest directly at the data layer to safeguard all confidential files.

3. Security Advantage

Using a layered approach centralizes authentication, authorization, and data-access controls within specific application and data tiers. This ensures that sensitive student information and placement documents can only be accessed by authorized users.

4. Performance Benefit

During peak placement season, hundreds of students access the portal simultaneously to apply for drives. Layering our application allows us to isolate and optimize high-load tasks—such as batch eligibility filtering —inside the business layer. This ensures the system easily supports over 2,000 concurrent users while maintaining response times under 3 seconds.